THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 2	$2250 = 2 \times 3^2 \times 5^3$
2	**(c)** 1	Each point where the graph meets the x-axis is a zero
3	**(d)** infinitely many solutions (coincident lines)	Compare a_1/a_2, b_1/b_2, c_1/c_2
4	**(b)** 7	Sum $= -b/a$
5	**(a)** 22338	LCM $= \dfrac{306 \times 657}{9}$
6	**(c)** 5	Sum of zeroes $= 7$, so other $= 7 - 2 = 5$
7	**(b)** 5	Solve by elimination
8	**(a)** 9	$D = 0$: $4m^2 - 4k = 0$
9	**(a)** always irrational	If it were rational, dividing by the rational would make the irrational rational
10	**(d)** $x^2 - 7x + 12$	$x^2 - (\text{sum})x + \text{product}$
11	**(b)** 2	Solve by elimination
12	**(b)** 14	Other root 7; product $= k$
13	**(b)** 5	$3125 = 5^5$; the number of places is the larger of the powers of 2 and 5
14	**(b)** 9	$\alpha + \beta = -b/a$
15	**(d)** 24	$x = 6$, $y = 4$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$6 = 2 \times 3$ has no factor 5 (1); so 6^n never ends in 0 (1).	1+1
17	Zeroes are -2 and 3 (1); a parabola, so degree 2 (1).	1+1
18	Let father x, daughter y now (1); $x - 7 = 7(y - 7)$ and $x + 3 = 3(y + 3)$ (1).	1+1
19	Integers $x, x+1$: $x(x+1) = 72$ (1); $x^2 + x - 72 = 0$ (1).	1+1
20	LCM $= \dfrac{306 \times 657}{9}$ (1) $= 22338$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Assume $\sqrt{2} = a/b$ in lowest terms, $b \neq 0$ (1); then $2b^2 = a^2$, so $2 \mid a^2$ hence $2 \mid a$ (1); put $a = 2c$: $b^2 = 2c^2$ so $2 \mid b$ - contradicts coprime (1).	1+1+1
22	Zeroes -2, 3 (1); sum $1 = -b/a$ (1); product $-6 = c/a$ (1).	1+1+1
23	$\frac{a_1}{a_2} = \frac{5}{7}$, $\frac{b_1}{b_2} = -\frac{2}{3}$, $\frac{c_1}{c_2} = \frac{8}{9}$ (2); ratios $a_1/a_2 \neq b_1/b_2$: lines intersect (1).	2+1
24	$D = (-7)^2 - 4(2)(3) = 25$ (1); $x = \dfrac{7 \pm 5}{4}$ (1); roots $x = 3, \frac{1}{2}$ (1).	1+1+1
	OR: Assume $3 + 2\sqrt{5} = r$ rational (1); then $\sqrt{5} = \dfrac{r - 3}{2}$ is rational (1) - contradiction; so it is irrational (1).	1+1+1
25	$p(2) = 0$: $4 - 10 + 6 = 0$ gives $k = -5$ (1); sum of zeroes $= -k = 5$ (1); other zero $= 5 - (2) = 3$ (1).	1+1+1
	OR: Multiply (i) by 4 and (ii) by 3 and subtract: $(-7)x = -21$ (2); $x = 3$ (1); put in (i): $y = 2$ (1); check in (ii) (1).	2+1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Assume $\sqrt{17} = a/b$, with a, b coprime integers, $b \neq 0$ (1); squaring, $17b^2 = a^2$, so $17 \mid a^2$ and hence $17 \mid a$ (1); write $a = 17c$: $b^2 = 17c^2$, so $17 \mid b$ (1); then 17 is a common factor of a and b, contradicting coprimality (1); so $\sqrt{17}$ is irrational (1).	1+1+1+1+1

	OR: $\alpha+\beta=7$, $\alpha\beta=10$ (1+1); new sum $=9$ (1); new product $=\alpha\beta+(\alpha+\beta)+1=18$ (1); polynomial $x^2-(9)x+(18)$ (1).	1+1+1+1+1
27	Son x, father $8x$ (1); $8x+9=5(x+9)$ (2); $x=12$ (1); son 12 years, father 96 years (1).	1+2+1+1
	OR: Speed x: $\frac{360}{x}-\frac{360}{x+5}=1$ (1); $1800=1x(x+5)$ so $x^2+5x-1800=0$ (2); $(x-40)(x+45)=0$ (1); speed 40 km/h (1).	1+2+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 3 and 5	Factorise
2	**(d)** $\sqrt{5}+3$	Rational + irrational is irrational
3	**(b)** 14	$\alpha\beta = c/a$
4	**(d)** 4	$\frac{1}{2}=\frac{2}{k}$ e$\frac{3}{7}$
5	**(b)** 25	$D = b^2 - 4ac$
6	**(b)** no natural number n	$3 = 3$ lacks the prime factor 5
7	**(c)** 0	2 is a zero, so the remainder is 0
8	**(d)** 2	Solve, then add
9	**(d)** two distinct real roots	$D = b^2 - 4ac = 12$
10	**(a)** 5	$480 = 2^5 \times 3 \times 5$
11	**(c)** 7	Sum $= -b/a$
12	**(d)** no solution (parallel lines)	Compare a_1/a_2, b_1/b_2, c_1/c_2
13	**(c)** 9	Sum $= -b/a$
14	**(d)** 2	$3600 = 2^4 \times 3^2 \times 5^2$
15	**(b)** 2	Each point where the graph meets the x-axis is a zero

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Sum $= -\frac{b}{a} = 4$ (1); product $= \frac{c}{a} = 3$ (1).	1+1
17	Express one variable, substitute: $x = 5$ (1), $y = 3$ (1).	1+1
18	Split the middle term and factorise (1); roots $\frac{3}{2}, 2$ (1).	1+1
19	$7 \times 6 + 7 = 7(6+1)$ (1) has factors 7 and 7, other than 1 and itself (1).	1+1
20	$x^2 - (\frac{1}{4})x + (-1)$ (1) $= x^2 - \frac{1}{4}x - 1$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Divide by 1: $x^2 - 6x + 8 = 0$ (1); $(x-3)^2 = 1$ (1); $x - 3 = \pm 1$, so $x = 2, 4$ (1).	1+1+1
22	$26 = 2 \times 13$, $91 = 7 \times 13$ (1); HCF $= 13$, LCM $= 182$ (1); $13 \times 182 = 2366$ verified (1).	1+1+1
23	Sum $= 10$, product $= 21$ (1+1); polynomial $x^2 - (10)x + (21) = x^2 - 10x + 21$ (1).	1+1+1
24	Let tens digit x, ones digit y: $x + y = 9$ (1); $(10y + x) - (10x + y) = -45$ so $y - x = -5$ (1); $x = 7$, $y = 2$, number 72 (1).	1+1+1
	OR: $D = k^2 - 4(4) = 0$ (1); $k^2 = 16$ (1); $k = \pm 4$ (1).	1+1+1
25	Assume $\sqrt{3} = a/b$ in lowest terms, $b \neq 0$ (1); then $3b^2 = a^2$, so $3 \mid a^2$ hence $3 \mid a$ (1); put $a = 3c$: $b^2 = 3c^2$ so $3 \mid b$ - contradicts coprime (1).	1+1+1
	OR: Zeroes $\frac{1}{2}$, 4 (1); sum $\frac{9}{2} = -b/a$ (1); product $2 = c/a$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Maximum weight $=$ HCF$(420, 130) = 10$ kg (3); trays $= \frac{420}{10} + \frac{130}{10} = 42 + 13 = 55$ (2).	3+2
	OR: Sum of zeroes $= -b/a = 8$ (2); third zero $= 8 - (1) - (2) = 5$ (2); check by product $= 10$ (1).	2+2+1
27	Fixed x, per km y: $x + 10y = 170$, $x + 15y = 240$ (2); subtract: $5y = 70$, $y = 14$ (1); $x = 30$ (1); fare for 25 km $= 30 + 14 \times 25 = 380$ (1).	2+1+1+1

	OR: Breadth x, length $x+7$: $x(x+7)=330$ (1); $x^2+7x-330=0$ (1); $(x-15)(x+22)=0$ (1); $x=15$ m (breadth), length 22 m (reject negative) (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** 4	Solve by elimination
2	**(d)** 16	$D = 0$: $4m^2 - 4k = 0$
3	**(a)** 23460	LCM $= \dfrac{510 \times 92}{2}$
4	**(a)** 6	Sum of zeroes $= 9$, so other $= 9 - 3 = 6$
5	**(b)** 1	Solve by elimination
6	**(d)** 24	Other root 8; product $= k$
7	**(c)** always irrational	If it were rational, dividing by the rational would make the irrational rational
8	**(c)** $x^2 - 3x - 10$	$x^2 - (\text{sum})x + \text{product}$
9	**(d)** 45	$x = 9$, $y = 5$
10	**(d)** -4 and 6	Factorise
11	**(c)** 3	$8 = 2^3$; the number of places is the larger of the powers of 2 and 5
12	**(a)** 5	$\alpha + \beta = -b/a$
13	**(d)** 9	$\frac{1}{3}=\frac{3}{k}$ e$\frac{4}{13}$
14	**(a)** -56	$D = b^2 - 4ac$
15	**(c)** $\sqrt{7} + 2$	Rational + irrational is irrational

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$\frac{1}{2} = \frac{2}{k} \neq \frac{3}{7}$ (1); $k = 4$ (1).	1+1
17	$D = 0$: $(8)^2 - 4k = 0$ (1); $k = 16$ (1).	1+1
18	$14 = 2 \times 7$ has no factor 5 (1); so 14^n never ends in 0 (1).	1+1
19	Zeroes are 1 and 4 (1); a parabola, so degree 2 (1).	1+1
20	Let father x, daughter y now (1); $x - 7 = 6(y - 7)$ and $x + 3 = 3(y + 3)$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$\frac{a_1}{a_2} = \frac{1}{2}$, $\frac{b_1}{b_2} = \frac{1}{2}$, $\frac{c_1}{c_2} = \frac{1}{2}$ (2); all ratios equal: lines coincide (1).	2+1
22	$D = (-6)^2 - 4(1)(4) = 20 > 0$ (1); $x = \dfrac{6 \pm \sqrt{20}}{2}$ (1); roots $\dfrac{6 + \sqrt{20}}{2}$ and $\dfrac{6 - \sqrt{20}}{2}$ (1).	1+1+1
23	Assume $2 + 3\sqrt{7} = r$ rational (1); then $\sqrt{7} = \dfrac{r - 2}{3}$ is rational (1) - contradiction; so it is irrational (1).	1+1+1
24	$p(3) = 0$: $9 - 21 + 12 = 0$ gives $k = -7$ (1); sum of zeroes $= -k = 7$ (1); other zero $= 7 - (3) = 4$ (1).	1+1+1
	OR: Multiply (i) by -3 and (ii) by 2 and subtract: $(-13)x = -52$ (2); $x = 4$ (1); put in (i): $y = 1$ (1); check in (ii) (1).	2+1+1+1
25	Divide by 1: $x^2 - 2x - 15 = 0$ (1); $(x - 1)^2 = 16$ (1); $x - 1 = \pm 4$, so $x = -3, 5$ (1).	1+1+1
	OR: $510 = 2 \times 3 \times 5 \times 17$, $92 = 2^2 \times 23$ (1); HCF $= 2$, LCM $= 23460$ (1); $2 \times 23460 = 46920$ verified (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Assume $\sqrt{19} = a/b$, with a, b coprime integers, $b \neq 0$ (1); squaring, $19b^2 = a^2$, so $19 \mid a^2$ and hence $19 \mid a$ (1); write $a = 19c$: $b^2 = 19c^2$, so $19 \mid b$ (1); then 19 is a common factor of a and b, contradicting coprimality (1); so $\sqrt{19}$ is irrational (1).	1+1+1+1+1
	OR: $\alpha + \beta = 3$, $\alpha\beta = -4$ (1+1); new sum $= 5$ (1); new product $= \alpha\beta + (\alpha + \beta) + 1 = 0$ (1); polynomial $x^2 - (5)x + (0)$ (1).	1+1+1+1+1

27	Son x, father $7x$ (1); $7x + 6 = 4(x + 6)$ (2); $x = 6$ (1); son 6 years, father 42 years (1).	1+2+1+1
	OR: Speed x: $\frac{420}{x} - \frac{420}{x+10} = 1$ (1); $4200 = 1x(x + 10)$ so $x^2 + 10x - 4200 = 0$ (2); $(x - 60)(x + 70) = 0$ (1); speed 60 km/h (1).	1+2+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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